Bagging on decision trees

A worked example on twelve training runs, showing where every number comes from

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Notation and colour code

One small dataset (twelve past training runs) and one question: *what happens if, instead of growing one decision tree, we grow many on resampled data and average them?* Every step is computed by hand, with a calculator.

Colours mark provenance. Every number that comes from a particular tree is shaded in that tree's colour: tree 1 (sample S1) tree 2 (sample S2) tree 3 (sample S3). A number with no shading is shared by all trees or belongs to the original data. In the tree pictures, the **yellow path with thick arrows** is the route a new run follows, and the leaf it reaches is where the number is read.

Six ideas you need first

Features and label. Each run has three recorded features (learning rate, batch size, warm-up yes/no) and an outcome: **D** (diverged, the loss blew up) or **C** (converged). **Decision tree.** Asks a yes/no question about one feature, then another, until each group of runs is pure. **Entropy.** For a group in which a fraction p of runs diverged, $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$: 0 for a pure group, 1 for a 50/50 group. **Gain.** Entropy before a question minus the size-weighted entropy after it. A tree asks, at every group, the question of largest gain. **Bootstrap sample.** Draw as many runs as the dataset has, with replacement. **Variance, covariance, correlation.** $\mathrm{Var}(X) = E[(X-EX)^2]$; $\mathrm{Cov}(X,Y) = E[(X-EX)(Y-EY)]$; $\rho = \mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$, between -1 and 1.

1 The twelve runs and the single tree

run	rate	batch	warm-up	out	run	rate	batch	warm-up	out
1	0.3	256	yes	**D**	7	0.03	256	no	**C**
2	0.3	32	no	**D**	8	0.03	64	yes	**C**
3	0.1	128	yes	**D**	9	0.01	128	yes	**C**
4	0.1	64	no	**D**	10	0.01	32	yes	**C**
5	0.03	32	no	**D**	11	0.001	64	no	**C**
6	0.01	32	no	**D**	12	0.001	256	yes	**C**

Grown on all twelve runs, the tree asks: rate ≤ 0.065? If no, predict **D** (runs 1–4). If yes: batch ≤ 48? If no, predict **C** (runs 7, 8, 9, 11, 12). If yes (runs 5, 6, 10): warm-up? No $\rightarrow$ **D**; yes $\rightarrow$ **C**.

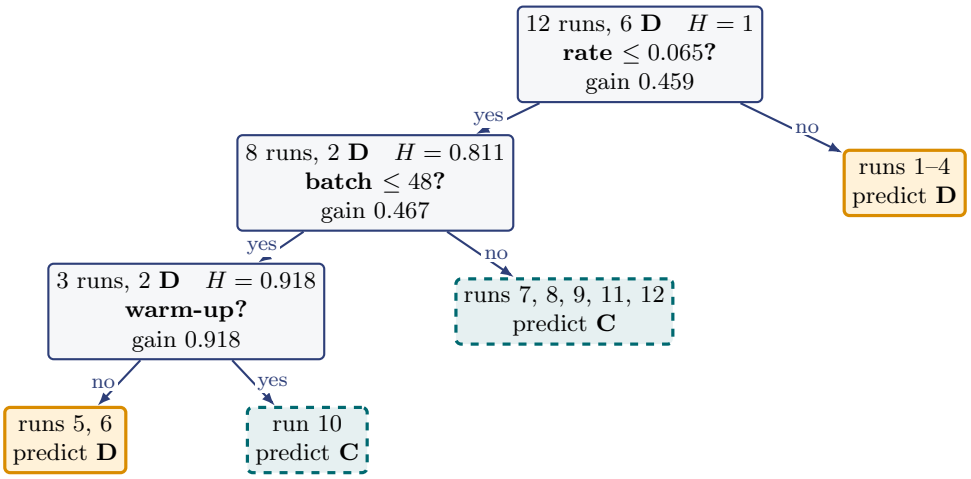


Figure 1: The single tree grown from all twelve runs. "Yes" goes left, "no" goes right.

Two new runs that every method below will predict:

new run	rate	batch	warm-up	the single tree says
A	0.05	32	no	**D** (rate ≤ 0.065 yes; batch ≤ 48 yes; warm-up no)
B	0.2	64	yes	**D** (rate ≤ 0.065 no)

The single tree is confident about both. Bagging will show how much that confidence is worth.

2 Step 1: draw three bootstrap samples

A **bootstrap sample** takes 12 draws from the 12 runs; after each draw the run goes back, so a run may be drawn 0, 1, 2 or 3 times. The table shows, for every run, how many times each sample drew it (darker shading = more draws).

run	rate	batch	warm-up	out	S1	S2	S3
1	0.3	256	yes	**D**	3	0	0
2	0.3	32	no	**D**	0	2	0
3	0.1	128	yes	**D**	1	0	1
4	0.1	64	no	**D**	2	2	0
5	0.03	32	no	**D**	0	2	3
6	0.01	32	no	**D**	0	1	2
7	0.03	256	no	**C**	1	0	2
8	0.03	64	yes	**C**	2	1	2
9	0.01	128	yes	**C**	0	1	0
10	0.01	32	yes	**C**	1	2	0
11	0.001	64	no	**C**	2	1	0
12	0.001	256	yes	**C**	0	0	2
draws in the sample (row sums)					12	12	12

Sample S1 never drew runs 2, 5, 6, 9, 12; S2 never drew 1, 3, 7, 12; S3 never drew 1, 2, 4, 9, 10, 11. Each sample is a slightly different picture of "what past runs looked like". On average a sample holds about 63% of the distinct runs (each run is missed with probability $(1 - \frac{1}{12})^{12} = 0.352$, and about $1/e = 0.368$ for a large dataset).

3 Step 2: grow one full tree on each sample

Each tree is grown by the same rule as the single tree (largest entropy gain, stop when pure). Only the data differ. The number each tree *uses* later is the fraction of **D** draws in the leaf a new run reaches;

here every leaf is pure, so it is 0 or 1.

Tree 1, from sample S1

S1 holds 6 **D** draws (runs 1, 1, 1, 3, 4, 4) and 6 **C** draws (runs 7, 8, 8, 10, 11, 11), so $H = 1$. The question **rate** ≤ 0.065 sends runs 7, 8, 8, 10, 11, 11 one way (6 draws, no **D**) and runs 1, 1, 1, 3, 4, 4 the other (6 draws, all **D**): both groups are pure, so

$$\text{gain} = 1 - \left(\tfrac{6}{12} \cdot 0 + \tfrac{6}{12} \cdot 0\right) = 1.0000 .$$

This is the largest possible gain, so tree 1 stops after one question.

Tree 2, from sample S2

S2 holds 7 **D** draws (runs 2, 2, 4, 4, 5, 5, 6) and 5 **C** draws (8, 9, 10, 10, 11): $H(\tfrac{7}{12}) = 0.9799$. The best root question is **warm-up** $=$ **no**. It sends runs 2, 2, 4, 4, 5, 5, 6, 11 one way (8 draws, 7 **D**, $H(\tfrac{7}{8}) = 0.5436$) and runs 8, 9, 10, 10 the other (4 draws, all **C**, $H = 0$):

$$\text{gain} = 0.9799 - \left(\tfrac{8}{12} \times 0.5436 + \tfrac{4}{12} \times 0\right) = 0.9799 - 0.3624 = 0.6175 .$$

The 8-draw group is not pure (run 11 is **C**), so tree 2 asks a second question there: **rate** ≤ 0.0055 (halfway between 0.001 and 0.01) isolates run 11 alone, gain $H(\tfrac{7}{8}) = 0.5436$. Runner-up root questions score much less (rate ≤ 0.065 gives 0.344): S2 lacks runs 1 and 3, the two **D** runs *with* warm-up, so within S2 warm-up looks like the cause.

Tree 3, from sample S3

S3 holds 6 **D** draws (runs 3, 5, 5, 5, 6, 6) and 6 **C** draws (7, 7, 8, 8, 12, 12): $H = 1$. The best root question is **batch** ≤ 48. It sends runs 5, 5, 5, 6, 6 one way (5 draws, all **D**, $H = 0$) and runs 3, 7, 7, 8, 8, 12, 12 the other (7 draws, 1 **D**, $H(\tfrac{1}{7}) = 0.5917$):

$$\text{gain} = 1 - \left(\tfrac{5}{12} \times 0 + \tfrac{7}{12} \times 0.5917\right) = 1 - 0.3451 = 0.6549 .$$

The 7-draw group holds run 3 (rate 0.1, **D**) and six **C** draws with rate ≤ 0.03, so a second question **rate** ≤ 0.065 splits them perfectly (gain 0.5917).

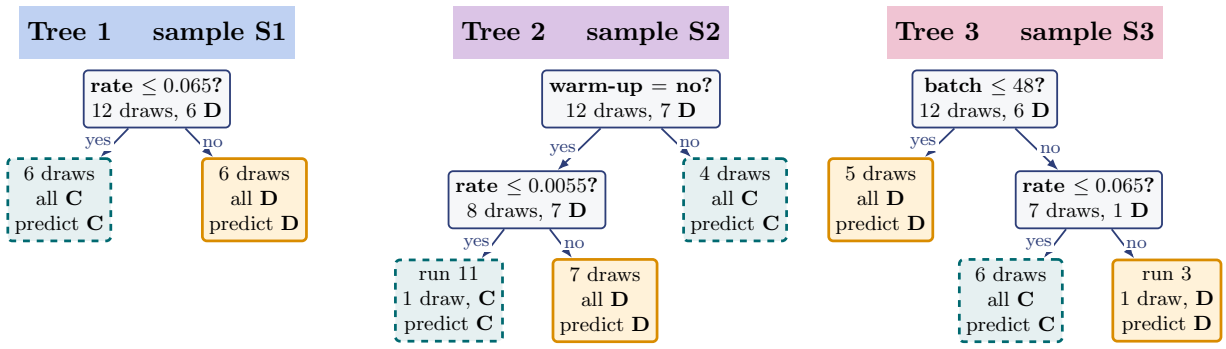


Figure 2: The three trees, grown from three bootstrap samples of the same twelve runs. They differ in the first question, in the second question, and in the thresholds (rate ≤ 0.0055 in tree 2, rate ≤ 0.065 in trees 1 and 3), but all three use the learning rate.

4 Step 3: send a new run through every tree, and read one number from each

The number we take from tree b is $T_b(x)$: the fraction of **D** draws in the leaf the run reaches (1 if the leaf predicts **D**, 0 if **C**, since these leaves are pure).

New run A: rate 0.05, batch 32, warm-up no

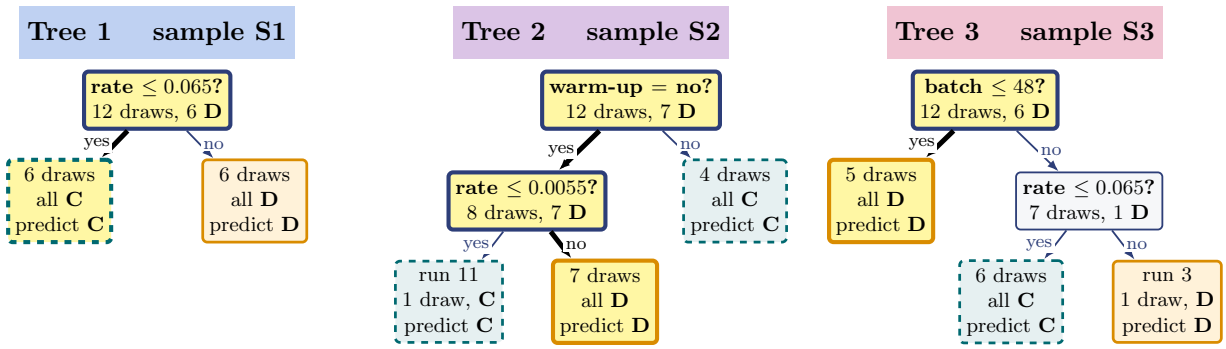


Figure 3: The path of run A (yellow, thick arrows). Tree 1: rate $0.05 \le 0.065$, yes → leaf "6 draws, all **C**". Tree 2: warm-up is no, yes; rate $0.05 > 0.0055$, no → leaf "7 draws, all **D**". Tree 3: batch $32 \le 48$, yes → leaf "5 draws, all **D**".

tree	leaf reached	fraction of **D** draws in it	$T_b(A)$
1 (S1)	rate ≤ 0.065: yes	0/6	**0**
2 (S2)	warm-up no; rate > 0.0055	7/7	**1**
3 (S3)	batch ≤ 48: yes	5/5	**1**

New run B: rate 0.2, batch 64, warm-up yes

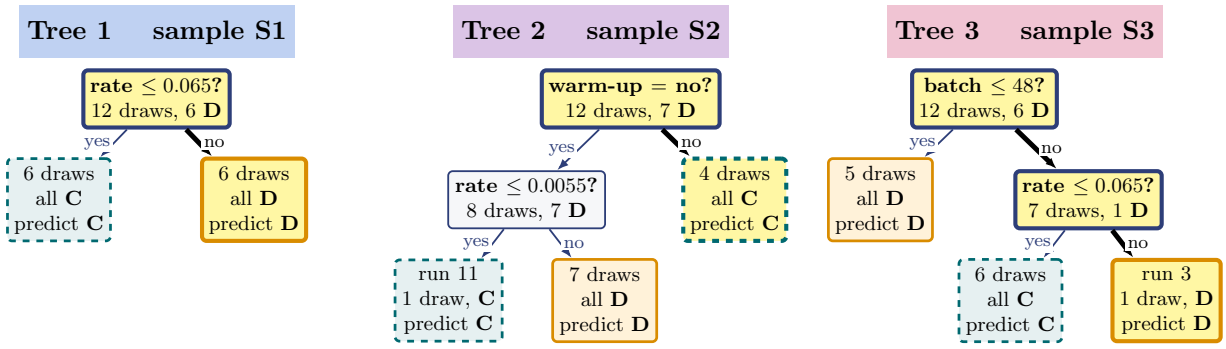


Figure 4: The path of run B. Tree 1: rate $0.2 > 0.065$, no → "6 draws, all **D**". Tree 2: warm-up is yes, so the answer is "no" to "warm-up = no" → "4 draws, all **C**". Tree 3: batch $64 > 48$, no; rate $0.2 > 0.065$, no → "run 3, **D**".

tree	leaf reached	fraction of **D** draws in it	$T_b(B)$
1 (S1)	rate ≤ 0.065: no	6/6	**1**
2 (S2)	warm-up yes (so not "no")	0/4	**0**
3 (S3)	batch > 48; rate > 0.065	1/1	**1**

Tree 2 answers **C** for run B because its sample has no high-rate run that used warm-up. That is the disagreement bagging is designed to average over.

5 Step 4: average the numbers, then threshold at $\frac{1}{2}$

The bagged prediction is $\bar{T} = \frac{1}{B}\sum_b T_b$, and the bagged answer is **D** when $\bar{T} > \frac{1}{2}$.

$$\text{run A: } \bar{T} = \tfrac{1}{3}(\,0 + 1 + 1\,) = \tfrac{2}{3} \;\Rightarrow\; \mathbf{D}, \qquad \text{run B: } \bar{T} = \tfrac{1}{3}(\,1 + 0 + 1\,) = \tfrac{2}{3} \;\Rightarrow\; \mathbf{D}.$$

Both answers agree with the single tree, but the single tree showed no doubt, and the vote shows that one tree in three disagrees on each run. **Averaging turned a hidden disagreement into a number.**

Three trees give a rough estimate: more trees change the number

With only three trees, $\bar{T}$ can only be $0, \frac{1}{3}, \frac{2}{3}, 1$. When we draw many more bootstrap samples (a computer simulation, 20,000 trees), the fraction of trees that say **D** at run A settles at 0.872 and at run B at 0.934. Three trees gave $\frac{2}{3} = 0.667$ for both: the same answer, but a rough estimate of the confidence. With 200 trees the same simulation gives 0.845 and 0.965.

6 Why averaging helps: the variance, derived step by step

Setup. Fix one new run x. Let $T_1, \ldots, T_B$ be the predictions at x of B trees, each with variance σ^2, and suppose every pair has the same correlation ρ, so $\mathrm{Cov}(T_b, T_c) = \rho\sigma^2$ for $b \neq c$. Goal: $\mathrm{Var}(\bar{T})$ for $\bar{T} = \frac{1}{B}\sum_b T_b$.

Step 1. *Pull out the constant.* A constant a multiplies a variance by a^2: $\mathrm{Var}(\bar{T}) = \frac{1}{B^2}\mathrm{Var}\big(\sum_b T_b\big)$.

Step 2. *Expand the sum.* Every pair of trees appears: $\mathrm{Var}\big(\sum_b T_b\big) = \sum_{b=1}^{B}\sum_{c=1}^{B}\mathrm{Cov}(T_b, T_c)$.

Step 3. *Count the terms.* B terms with $b = c$, each σ^2; $B(B-1)$ terms with $b \neq c$, each $\rho\sigma^2$. So the sum is $B\sigma^2 + B(B-1)\rho\sigma^2$.

Step 4. *Divide by B^2.* $\mathrm{Var}(\bar{T}) = \dfrac{\sigma^2}{B} + \dfrac{B-1}{B}\rho\sigma^2 = \rho\sigma^2 + \dfrac{\sigma^2}{B} - \dfrac{\rho\sigma^2}{B} = \boxed{\rho\sigma^2 + (1-\rho)\dfrac{\sigma^2}{B}}$.

Step 5. *Read it.* $\rho = 0$: σ^2/B. $\rho = 1$: σ^2, no gain. $B \to \infty$: the floor $\rho\sigma^2$ remains.

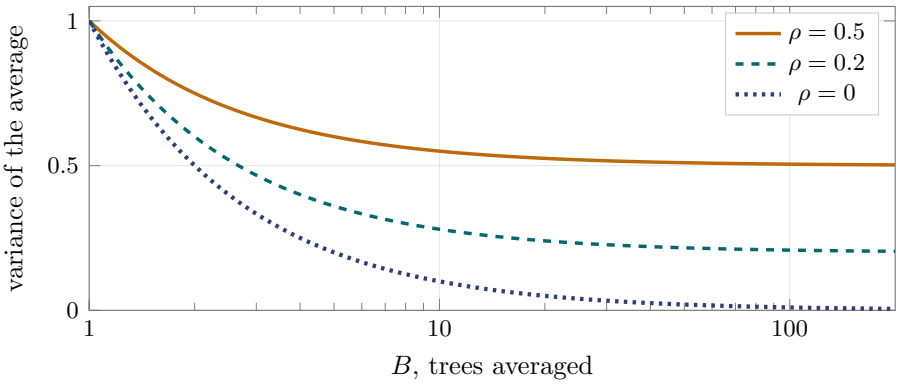


Figure 5: $\sigma^2 = 1$. With $\rho = 0.5$: $B = 1$: 1; $B = 10$: 0.55; $B = 100$: 0.505; floor 0.5. Each curve flattens at its floor $\rho\sigma^2$.

Checking the formula on our trees

Resample the twelve runs, grow a tree, record its prediction at run A, and repeat many times. At run A a tree says **D** with probability $p = 0.872$, so one tree's variance over the resampling is $\sigma^2_{\text{boot}} = p(1-p) = 0.1115$. The table compares the variance of the average of B trees, measured directly by simulation (4000 repeats each), with σ^2_{boot}/B:

B trees averaged	measured variance of the average	σ^2_{boot}/B
1	0.1118	0.1115
3	0.0362	0.0372
10	0.0111	0.0111
50	0.0021	0.0022

They agree. This confirms the σ^2/B part of the formula for trees drawn from the *same* twelve runs. It does not show a floor, and that is the point of the next paragraph.

Where the correlation ρ comes from

Trees grown by resampling the same data are independent of each other *given the data* D, so their variance shrinks like $1/B$, as the table shows. But the data themselves are one random draw from all the runs that could have happened, and *all* the trees share it. Write $\mu(D) = E[T \mid D]$ for the average prediction that the resampling would produce from dataset D. For two different trees $b \neq c$ grown from the same D:

$$\operatorname{Cov}(T_b, T_c) = \underbrace{E\big[\operatorname{Cov}(T_b, T_c \mid D)\big]}_{=0,\text{ independent given } D} + \operatorname{Cov}\big(\mu(D), \mu(D)\big) = \operatorname{Var}\big(\mu(D)\big),$$

and one tree's variance splits as $\sigma^2 = \operatorname{Var}(\mu(D)) + E[\operatorname{Var}(T \mid D)]$. So

$$\rho\sigma^2 = \operatorname{Var}\big(\mu(D)\big) \text{ (the part due to the data)}, \qquad (1-\rho)\sigma^2 = E\big[\operatorname{Var}(T \mid D)\big] \text{ (the part due to resampling)}.$$

In words: **averaging removes the variation due to resampling but cannot remove the variation of the dataset itself**, and that is the floor. With twelve runs we cannot measure $\operatorname{Var}(\mu(D))$ because we have only one dataset. On a larger simulated example with 200 runs and many fresh datasets, the measured values were $\sigma^2 = 0.071$, $\rho = 0.26$, so the floor was $\rho\sigma^2 = 0.019$, and the validation error of a single full tree, 0.144, fell to 0.115 for a 200-tree bag.

7 Do the trees have to use different features?

No. In plain bagging every split may use *any* feature, so the trees often share features. Above, all three trees use the learning rate, with different thresholds and order. Simulating 20,000 bootstrap trees of the twelve runs and recording the feature of the **root** question:

	rate	batch	warm-up
bagging (all 3 features allowed at every split)	80.8%	12.2%	6.9%
random forest, 2 random features per split	58.7%	28.0%	13.2%
random forest, 1 random feature per split	35.2%	33.2%	31.6%

Plain bagging starts with the learning rate in about four trees out of five, so its trees are alike: ρ is high and the floor $\rho\sigma^2$ is high. A **random forest** lets each split see only a random subset of the features. Trees that are not offered the learning rate must start with batch size or warm-up, so the trees differ more, ρ falls, and so does the floor. The price is that each individual tree is a little worse (a larger σ^2): on the 200-run simulation σ^2 rose from 0.071 to 0.082 while ρ fell from 0.26 to 0.20, so the floor fell from 0.019 to 0.016.

8 The out-of-bag error

A run left out of a tree's sample is a fresh test run for that tree. For every training run, average only the trees that did *not* draw it, and compare with its true outcome. With $B = 200$ trees (simulation), each run was out of bag in about 70 trees on average.

run	true	trees not drawing it	fraction saying **D**	OOB answer	right?
1	**D**	63	0.746	**D**	yes
2	**D**	65	0.985	**D**	yes
3	**D**	68	0.632	**D**	yes
4	**D**	77	0.818	**D**	yes
5	**D**	70	0.557	**D**	yes
6	**D**	74	0.189	**C**	**no**
7	**C**	69	0.406	**C**	yes
8	**C**	66	0.455	**C**	yes
9	**C**	80	0.037	**C**	yes
10	**C**	69	0.783	**D**	**no**
11	**C**	70	0.514	**D**	**no**
12	**C**	68	0.000	**C**	yes

Out-of-bag error: 3 wrong out of 12, that is 0.25. The single tree, checked on the runs it was grown from, makes 0 errors: a tree that fits its own data is an optimistic estimate. The out-of-bag figure is honest, and it costs nothing extra: no runs were held back.

9 Summary

step	what you do	which number is used
1. resample	draw 12 runs with replacement, B times	the counts of draws per run (Step 1 table)
2. grow	full tree on each sample, largest entropy gain, stop when pure	each tree's own questions and leaf contents
3. ask	send the new run down every tree	T_1, T_2, T_3: the **D**-fraction of the leaf reached in each tree
4. average	$\bar{T} = \frac{1}{B}\sum_b T_b$; predict **D** if $\bar{T} > \frac{1}{2}$	the three numbers, added and divided by B
why it works	$\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$	σ^2: one tree's variance; ρ: shared-data correlation
what it cannot fix	the floor $\rho\sigma^2$; the bias of a tree	lower ρ: random forest

10 Review questions

1. In sample S2, why is warm-up the best first question when in the full data it is not? *Answer:* S2 does not contain runs 1 and 3, the two **D** runs that used warm-up.
2. Which number do we take from tree 2 for run B, and why? *Answer:* $T_2(B) = 0$: warm-up is yes, so the run goes to the leaf "4 draws, all **C**".
3. With $\sigma^2 = 1$, $\rho = 0.3$, $B = 20$: the variance of the average, and the floor? *Answer:* $0.3 + 0.7/20 = 0.335$; floor 0.3.
4. Why does the variance table match σ^2/B but show no floor? *Answer:* the trees were grown from the same twelve runs, so only resampling varies; the floor comes from the dataset varying, which we cannot see from one dataset.
5. Why is a random forest's ρ lower than bagging's? *Answer:* each split sees only some features, so trees are forced to use different questions, as the root-question table shows.